

Exercise 4: Emerging Applications in Retail and E-Commerce

Retail and e-commerce are increasingly shaped by systems that can interpret visual and audio information in real time. Computer vision has moved beyond basic image classification and is now used to detect products, monitor shelves, analyse store activity and support more personalised digital shopping experiences. Audio processing adds another useful layer by enabling speech interaction, recognising acoustic events and extracting vocal features linked to user intent or frustration. Together, these technologies are helping retailers reduce friction, improve operational efficiency and create more adaptive customer experiences. This wider shift matters because retail competition is no longer based only on price and product availability, but also on convenience, responsiveness and service quality (Grewal, Roggeveen and Nordfält, 2017).

One emerging application is **visual product search**. In many online retail settings, customers know what they want when they see it, but struggle to describe it accurately in words. Computer vision helps solve this by allowing users to upload an image and retrieve visually similar products. This is especially valuable in fashion, home décor and beauty, where visual style strongly affects purchase decisions. Recent e-commerce research shows that vision-language pre-training can improve retrieval systems by combining image information with product text, creating stronger matching between customer queries and catalogue items (Zheng *et al.*, 2023). This matters commercially because it shortens the path from inspiration to purchase and can reduce search abandonment. However, the quality of visual search still depends on strong catalogue imagery, robust embeddings and good handling of cluttered backgrounds, occlusion and lighting variation in user-provided images. In practice, the challenge is not proving that visual search works in principle, but making it reliable across huge, messy and constantly changing product inventories (Zheng *et al.*, 2023).

A second major application is **automated shelf monitoring and on-shelf availability analysis**. In physical retail, empty shelves, misplaced products and poor planogram compliance can directly reduce sales. Computer vision helps by analysing camera feeds to detect missing or low-stock items and support faster replenishment. This turns shelf images into actionable stock information and improves operational efficiency. However, real retail environments remain challenging because packaging changes, products overlap, reflective materials distort images, and customers or trolleys may block the view. As a result, the main difficulty is maintaining reliable performance in live store conditions rather than simply achieving accuracy on controlled datasets (Higa and Iwamoto, 2019).

A third emerging application is **automated retail checkout**. Vision-based systems identify items directly from images or video, reducing reliance on barcode scanning. This can speed up checkout, reduce queues and create a smoother customer experience. However, it is also one of the most difficult retail vision tasks because systems must recognise similar-looking products, handle crowded scenes and link customer actions to the correct items. Privacy concerns also remain important, particularly when store-wide tracking is involved. For this reason, success depends not only on technical accuracy but also on trust and careful system design (Bukhari *et al.*, 2021).

A key computer vision technique relevant to these applications is **Mask R-CNN**. Mask R-CNN extends Faster R-CNN by adding a branch that predicts a pixel-level segmentation mask for each detected object, alongside the usual class label and bounding box (He *et al.*, 2017). Technically, the method first extracts features with a backbone convolutional network, then generates region proposals, and finally uses RoIAlign to preserve spatial precision before producing classification, localisation and segmentation outputs. This makes it especially useful in retail settings where simply detecting an object is not enough. For example, on crowded shelves or checkout trays, the

system may need to separate overlapping products and estimate the exact visible area of each one. The strength of Mask R-CNN is that it provides richer scene understanding than basic object detection alone. Its main weakness is speed. Because it uses region proposals and instance-level processing, it is usually heavier than faster one-stage approaches in real-time deployment scenarios (He *et al.*, 2017).

Another important state-of-the-art technique is **DETR (DEtection TRansformer)**. DETR is another important state-of-the-art technique. It uses a transformer encoder-decoder to detect objects directly as a set-prediction problem, without hand-designed steps like anchor generation or non-maximum suppression (Carion *et al.*, 2020). This allows stronger global reasoning across the whole image, which is useful in cluttered retail scenes such as shelves, baskets and busy store environments. Its main limitation is that the original model is more data-hungry and slower to train than earlier detectors, although later versions improved efficiency (Carion *et al.*, 2020).

Beyond mainstream applications, combining computer vision with audio processing can support more human-centred retail systems. One example is an **accessibility assistant for visually impaired shoppers**, where computer vision recognises aisles, labels and products, while audio processing enables speech input and spoken guidance. Audio features such as pitch, intensity and MFCCs can support robust speech interaction in retail environments (Issa, Demirci and Yazici, 2022). This is valuable because it improves accessibility rather than just efficiency, though the challenge is giving clear guidance in noisy and visually complex stores (Issa, Demirci and Yazici, 2022).

A second novel use is **self-checkout support through multimodal frustration detection**. Here, computer vision could detect failed scans, hesitation or item misplacement, while audio processing analyses vocal stress cues such as pitch change and speech energy. If both signals suggest difficulty, the system could simplify the interface or alert staff. This is useful because it links multimodal sensing to a real retail problem: checkout breakdown and customer frustration. Research suggests that combining multiple data types can improve predictive performance compared with single-modality systems (Zhang *et al.*, 2024).

In conclusion, computer vision is already transforming retail and e-commerce through applications such as visual search, shelf monitoring and automated checkout, while audio processing opens up more adaptive and inclusive forms of service. Mask R-CNN and DETR illustrate two important technical directions in this space: detailed instance-level understanding and end-to-end transformer-based detection. Looking ahead, the most valuable retail systems are likely to be multimodal, combining vision and audio in ways that solve practical problems rather than simply showcasing technical sophistication. The real challenge is not whether these systems can be built, but whether they can be deployed in ways that are accurate, usable, privacy-aware and genuinely beneficial to customers and staff.

References

- Bukhari, S.T., Amin, A.W., Naveed, M.A. and Abbas, M.R. (2021) 'ARC: A vision-based automatic retail checkout system', *arXiv*. Available at: arXiv:2104.02832.
- Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A. and Zagoruyko, S. (2020) 'End-to-end object detection with transformers', in *Computer Vision – ECCV 2020*. Cham: Springer, pp. 213–229.
- Grewal, D., Roggeveen, A.L. and Nordfält, J. (2017) 'The future of retailing', *Journal of Retailing*, 93(1), pp. 1–6.

He, K., Gkioxari, G., Dollár, P. and Girshick, R. (2017) 'Mask R-CNN', in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 2961–2969.

Higa, K. and Iwamoto, K. (2019) 'Robust shelf monitoring using supervised learning for improving on-shelf availability in retail stores', *Sensors*, 19(12), p. 2722.

Issa, D., Demirci, M.F. and Yazici, A. (2022) 'Speech emotion recognition with deep learning: a systematic review', *Results in Engineering*, 16, 100741.

Zhang, J., Wang, L., Wang, Z., Ma, Y. and Lu, B. (2024) 'Research on multimodal prediction of e-commerce customer satisfaction driven by big data', *Applied Sciences*, 14(18), p. 8181.

Zheng, X., Lv, F., Wang, Z., Liu, Q. and Zeng, X. (2023) 'Delving into e-commerce product retrieval with vision-language pre-training', *arXiv*. Available at: [arXiv:2304.04377](https://arxiv.org/abs/2304.04377).